# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 81.6 ± 1.96 |
| Correlation Stability | 71.1 ± 5.91 |
| Distribution Consistency | 82.4 ± 2.20 |
| Entropy Preservation | 90.8 ± 1.60 |
| Mean Z Score Drift | 0.2 ± 0.06 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 16.0 ± 0.00 |
| Data Density | 100.0 ± 0.00 |
| Coverage Score | 82.8 ± 0.40 |
| Feature Accuracy | 87.4 ± 4.45 |
| Novelty Score | 0.7 ± 0.02 |

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## Dataset Audit: breast_cancer_coimbra.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 116 |
| Columns | 10 |
| Numeric features | 10 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | gpt-4o-mini |
| Folds (*13) | 10 |
| Context % (*14) | 20% |
| Generation % (*15) | 100% |
| Join context (*16) | True |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| Age | 57.302 | 51.828 | -5.474 | 9.6% | OK |
| BMI | 27.582 | 27.952 | +0.370 | 1.3% | OK |
| Glucose | 97.793 | 102.699 | +4.906 | 5.0% | OK |
| Insulin | 10.012 | 8.671 | -1.341 | 13.4% | Warn |
| HOMA | 2.695 | 2.138 | -0.557 | 20.7% | Warn |
| Leptin | 26.615 | 19.859 | -6.756 | 25.4% | Drift |
| Adiponectin | 10.181 | 7.701 | -2.480 | 24.4% | Warn |
| Resistin | 14.726 | 12.103 | -2.623 | 17.8% | Warn |
| MCP.1 | 534.647 | 467.544 | -67.103 | 12.6% | Warn |
| Classification | 0.552 | 0.484 | -0.068 | 12.3% | Warn |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 81.6 ± 1.96 |
| Correlation Stability (*2) | 71.1 ± 5.91 |
| Distribution Consistency (*3) | 82.4 ± 2.20 |
| Entropy Preservation (*4) | 90.8 ± 1.60 |
| Mean Z Score Drift (*5) | 0.2 ± 0.06 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 16.0 ± 0.00 |
| Data Density (*8) | 100.0 ± 0.00 |
| Coverage Score (*9) | 82.8 ± 0.40 |

| | |
|---|---|
| Feature Accuracy (*10) | 87.4 ± 4.45 |
| Novelty Score (*11) | 0.7 ± 0.02 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7249999999999999 | 0.5875 | 0.6166666666666667 | -0.10833333333333335 | |
| F1 | 0.7310046394829003 | 0.5843843215667172 | 0.6489997099674519 | -0.08200492951544844 | |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5874999999999999 | 0.5416666666666667 | 0.6000000000000001 | 0.012499999999999994 | |
| F1 | 0.6191975641155285 | 0.5072874576472778 | 0.6541022565325971 | 0.03490469241706856 | |

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.4625 | 0.5583333333333333 | 0.5083333333333333 | 0.04583333333333332 | |
| F1 | 0.5361601301923884 | 0.5542614309844566 | 0.5352690675812986 | -0.00089106261108971 57 | |

### Model: LR

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7083333333333334 | 0.5916666666666666 | 0.7458333333333333 | 0.0375 | |
| F1 | 0.7179811943844927 | 0.6131286420281923 | 0.7496154602241558 | 0.03163426583966315 | |

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.4958333333333333 | 0.5291666666666667 | 0.5291666666666667 | 0.03333333333333326 | |
| F1 | 0.6547289527475286 | 0.481605401102225 | 0.5245231547840243 | -0.1302057979635043 | |

### Model: AB

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| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7458333333333333 | 0.65 | 0.75 | 0.00416666666666663 | |
| F1 | 0.754839950419102 | 0.641925983277357 | 0.7590429672041807 | 0.00420301678507870 56 | |

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.75 | 0.625 | 0.6875 | -0.06250000000000001 | |
| F1 | 0.7718283475783475 | 0.6395631081585473 | 0.710924381853917 | -0.0609039657244305 | |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7083333333333333 | 0.6416666666666668 | 0.7041666666666666 | -0.00416666666666685 | |
| F1 | 0.7331357451616072 | 0.6294209132918811 | 0.7291835445913407 | -0.00395220057026654 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6416666666666667 | 0.6249999999999999 | 0.6416666666666666 | 2.2204460492503132e-17 | |
| F1 | 0.5557236965932618 | 0.6584921200507938 | 0.5946269482151835 | 0.038903251621921695 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.725 | 0.6166666666666667 | 0.7249999999999999 | -2.2204460492503132e-17 | |
| F1 | 0.7359528619528619 | 0.612185468351797 | 0.741295566502463 | 0.005342704549601085 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 48.0 | 23.5 | 70.0 | 2.707 | 0.467408667 | 8.8071 | 9.7024 | 7.99585 | 417.114 | 0.0 |
| 2 | 83.0 | 20.69049454 | 92.0 | 3.115 | 0.706897333 | 8.8438 | 5.429285 | 4.06405 | 468.786 | 0.0 |
| 3 | 82.0 | 23.12467037 | 91.0 | 4.498 | 1.009651067 | 17.9393 | 22.43204 | 9.27715 | 554.697 | 0.0 |
| 4 | 68.0 | 21.36752137 | 77.0 | 3.226 | 0.612724933 | 9.8827 | 7.16956 | 12.766 | 928.22 | 0.0 |
| 5 | 86.0 | 21.11111111 | 92.0 | 3.549 | 0.8053864 | 6.6994 | 4.81924 | 10.57635 | 773.92 | 0.0 |
| 6 | 49.0 | 22.85445769 | 92.0 | 3.226 | 0.732086933 | 6.8317 | 13.67975 | 10.3176 | 530.41 | 0.0 |
| 7 | 89.0 | 22.7 | 77.0 | 4.69 | 0.890787333 | 6.964 | 5.589865 | 12.9361 | 1256.083 | 0.0 |
| 8 | 76.0 | 23.8 | 118.0 | 6.47 | 1.883201333 | 4.311 | 13.25132 | 5.1042 | 280.694 | 0.0 |
| 9 | 73.0 | 22.0 | 97.0 | 3.35 | 0.801543333 | 4.47 | 10.358725 | 6.28445 | 136.855 | 0.0 |
| 10 | 75.0 | 23.0 | 83.0 | 4.952 | 1.013839467 | 17.127 | 11.57899 | 7.0913 | 318.302 | 0.0 |
| 11 | 34.0 | 21.47 | 78.0 | 3.469 | 0.6674356 | 14.57 | 13.11 | 6.92 | 354.6 | 0.0 |
| 12 | 29.0 | 23.01 | 82.0 | 5.663 | 1.145436133 | 35.59 | 26.72 | 4.58 | 174.8 | 0.0 |
| 13 | 25.0 | 22.86 | 82.0 | 4.09 | 0.827270667 | 20.45 | 23.67 | 5.14 | 313.73 | 0.0 |
| 14 | 24.0 | 18.67 | 88.0 | 6.107 | 1.33 | 8.88 | 36.06 | 6.85 | 632.22 | 0.0 |
| 15 | 38.0 | 23.34 | 75.0 | 5.782 | 1.06967 | 15.26 | 17.95 | 9.35 | 165.02 | 0.0 |
| 16 | 44.0 | 20.76 | 86.0 | 7.553 | 1.6 | 14.09 | 20.32 | 7.64 | 63.61 | 0.0 |
| 17 | 47.0 | 22.03 | 84.0 | 2.869 | 0.59 | 26.65 | 38.04 | 3.32 | 191.72 | 0.0 |
| 18 | 61.0 | 32.03895937 | 85.0 | 18.077 | 3.790144333 | 30.7729 | 7.780255 | 13.68392 | 444.395 | 0.0 |
| 19 | 64.0 | 34.5297228 | 95.0 | 4.427 | 1.037393667 | 21.2117 | 5.46262 | 6.70188 | 252.449 | 0.0 |
| 20 | 32.0 | 36.51263743 | 87.0 | 14.026 | 3.0099796 | 49.3727 | 5.1 | 17.10223 | 588.46 | 0.0 |

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## B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 72.0 | 25.59 | 82.0 | 2.82 | 0.570392 | 24.96 | 33.75 | 3.27 | 392.46 | 1.0 |
| 2 | 69.0 | 35.09270153 | 101.0 | 5.646 | 1.4066068 | 83.4821 | 6.796985 | 82.1 | 263.499 | 0.0 |
| 3 | 76.0 | 29.2184076 | 83.0 | 5.376 | 1.1006464 | 28.562 | 7.36996 | 8.04375 | 698.789 | 0.0 |
| 4 | 48.0 | 28.125 | 90.0 | 2.54 | 0.56388 | 15.5325 | 10.22231 | 16.11032 | 1698.44 | 1.0 |
| 5 | 71.0 | 25.51020408 | 112.0 | 10.395 | 2.871792 | 19.0653 | 5.4861 | 42.7447 | 799.898 | 1.0 |
| 6 | 24.0 | 18.67 | 88.0 | 6.107 | 1.33 | 8.88 | 36.06 | 6.85 | 632.22 | 0.0 |
| 7 | 34.0 | 22.5 | 85.0 | 4.5 | 1.0 | 10.5 | 8.0 | 5.0 | 300.0 | 0.0 |
| 8 | 45.0 | 28.3 | 95.0 | 6.0 | 1.5 | 20.0 | 7.5 | 15.0 | 400.0 | 1.0 |
| 9 | 53.0 | 30.1 | 110.0 | 8.0 | 2.0 | 25.0 | 6.0 | 20.0 | 500.0 | 1.0 |
| 10 | 29.0 | 24.0 | 78.0 | 3.0 | 0.8 | 12.0 | 9.0 | 4.0 | 250.0 | 0.0 |
| 11 | 60.0 | 27.5 | 102.0 | 5.5 | 1.2 | 15.0 | 10.0 | 10.0 | 350.0 | 1.0 |
| 12 | 38.0 | 26.0 | 88.0 | 4.2 | 1.0 | 14.0 | 8.5 | 6.0 | 320.0 | 0.0 |
| 13 | 64.0 | 22.22222222 | 98.0 | 5.7 | 1.37788 | 12.1905 | 4.783985 | 13.91245 | 395.976 | 1.0 |
| 14 | 57.0 | 34.83814777 | 95.0 | 12.548 | 2.940414667 | 33.1612 | 2.36495 | 9.9542 | 655.834 | 1.0 |
| 15 | 68.0 | 21.08281329 | 102.0 | 6.2 | 1.55992 | 9.6994 | 8.574655 | 13.74244 | 448.799 | 1.0 |
| 16 | 42.0 | 21.35991456 | 93.0 | 2.999 | 0.6879706 | 19.0826 | 8.462915 | 17.37615 | 321.919 | 1.0 |

## B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 72.0 | 29.13631634 | 83.0 | 10.949 | 2.241625267 | 26.8081 | 2.78491 | 14.76966 | 232.018 | 1.0 |
| 2 | 35.0 | 35.2507611 | 90.0 | 6.817 | 1.513374 | 50.6094 | 6.966895 | 22.03703 | 667.928 | 0.0 |
| 3 | 66.0 | 26.5625 | 89.0 | 6.524 | 1.432235467 | 14.9084 | 8.42996 | 14.91922 | 269.487 | 1.0 |
| 4 | 71.0 | 30.3 | 102.0 | 8.34 | 2.098344 | 56.502 | 8.13 | 4.2989 | 200.976 | 0.0 |
| 5 | 61.0 | 32.03895937 | 85.0 | 18.077 | 3.790144333 | 30.7729 | 7.780255 | 13.68392 | 444.395 | 0.0 |
| 6 | 35.0 | 30.27681661 | 84.0 | 4.376 | 0.9067072 | 39.2134 | 9.048185 | 16.43706 | 733.797 | 0.0 |
| 7 | 54.0 | 28.5 | 88.0 | 7.2 | 1.6 | 30.5 | 8.1 | 15.2 | 300.5 | 0.0 |
| 8 | 45.0 | 31.2 | 95.0 | 11.5 | 2.5 | 40.3 | 6.8 | 18.5 | 400.1 | 1.0 |
| 9 | 67.0 | 26.8 | 92.0 | 5.8 | 1.2 | 20.1 | 9.5 | 12.3 | 250.7 | 0.0 |
| 10 | 73.0 | 29.9 | 100.0 | 9.0 | 2.0 | 35.2 | 7.4 | 14.8 | 350.2 | 1.0 |
| 11 | 58.0 | 33.5 | 85.0 | 6.0 | 1.4 | 25.7 | 10.2 | 17.0 | 280.3 | 0.0 |
| 12 | 62.0 | 27.3 | 91.0 | 8.4 | 1.9 | 28.9 | 5.6 | 16.1 | 320.9 | 1.0 |
| 13 | 50.0 | 38.57875854 | 106.0 | 6.703 | 1.752611067 | 46.6401 | 4.667645 | 11.78388 | 887.16 | 0.0 |
| 14 | 57.0 | 34.83814777 | 95.0 | 12.548 | 2.940414667 | 33.1612 | 2.36495 | 9.9542 | 655.834 | 1.0 |
| 15 | 38.0 | 22.4996371 | 95.0 | 5.261 | 1.232827667 | 8.438 | 4.77192 | 15.73606 | 199.055 | 1.0 |
| 16 | 81.0 | 31.64036818 | 100.0 | 9.669 | 2.38502 | 38.8066 | 10.636525 | 29.5583 | 426.175 | 1.0 |

## B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 49.0 | 29.77777778 | 70.0 | 8.396 | 1.449709333 | 51.3387 | 10.73174 | 20.76801 | 602.486 | 1.0 |
| 2 | 45.0 | 21.30394858 | 102.0 | 13.852 | 3.4851632 | 7.6476 | 21.056625 | 23.03408 | 552.444 | 1.0 |
| 3 | 73.0 | 22.0 | 97.0 | 3.35 | 0.801543333 | 4.47 | 10.358725 | 6.28445 | 136.855 | 0.0 |
| 4 | 48.0 | 32.46191136 | 99.0 | 28.677 | 7.0029234 | 46.076 | 21.57 | 10.15726 | 738.034 | 1.0 |
| 5 | 86.0 | 26.66666667 | 201.0 | 41.611 | 20.6307338 | 47.647 | 5.357135 | 24.3701 | 1698.44 | 1.0 |
| 6 | 32.0 | 36.51263743 | 87.0 | 14.026 | 3.0099796 | 49.3727 | 5.1 | 17.10223 | 588.46 | 0.0 |
| 7 | 34.0 | 24.5 | 85.0 | 6.5 | 1.5 | 15.2 | 10.5 | 12.3 | 200.5 | 0.0 |
| 8 | 52.0 | 30.1 | 110.0 | 15.0 | 3.0 | 22.0 | 8.0 | 18.0 | 450.0 | 1.0 |
| 9 | 41.0 | 27.3 | 95.0 | 9.5 | 2.2 | 18.5 | 12.0 | 14.5 | 320.0 | 0.0 |
| 10 | 60.0 | 31.8 | 130.0 | 20.0 | 4.5 | 30.0 | 6.5 | 25.0 | 600.0 | 1.0 |
| 11 | 29.0 | 22.0 | 75.0 | 5.0 | 1.0 | 10.0 | 15.0 | 8.0 | 150.0 | 0.0 |
| 12 | 47.0 | 28.5 | 105.0 | 12.0 | 2.8 | 25.0 | 9.5 | 20.0 | 400.0 | 1.0 |
| 13 | 43.0 | 31.25 | 103.0 | 4.328 | 1.099600533 | 25.7816 | 12.71896 | 38.6531 | 775.322 | 1.0 |
| 14 | 24.0 | 18.67 | 88.0 | 6.107 | 1.33 | 8.88 | 36.06 | 6.85 | 632.22 | 0.0 |
| 15 | 71.0 | 25.51020408 | 112.0 | 10.395 | 2.871792 | 19.0653 | 5.4861 | 42.7447 | 799.898 | 1.0 |
| 16 | 49.0 | 21.36752137 | 78.0 | 2.64 | 0.507936 | 6.3339 | 3.886145 | 22.94254 | 737.672 | 1.0 |

## B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 73.0 | 22.0 | 97.0 | 3.35 | 0.801543333 | 4.47 | 10.358725 | 6.28445 | 136.855 | 0.0 |
| 2 | 38.0 | 22.4996371 | 95.0 | 5.261 | 1.232827667 | 8.438 | 4.77192 | 15.73606 | 199.055 | 1.0 |
| 3 | 50.0 | 38.57875854 | 106.0 | 6.703 | 1.752611067 | 46.6401 | 4.667645 | 11.78388 | 887.16 | 0.0 |
| 4 | 45.0 | 37.03560819 | 83.0 | 6.76 | 1.383997333 | 39.9802 | 4.617125 | 8.70448 | 586.173 | 0.0 |
| 5 | 72.0 | 25.59 | 82.0 | 2.82 | 0.570392 | 24.96 | 33.75 | 3.27 | 392.46 | 1.0 |
| 6 | 29.0 | 32.27078777 | 84.0 | 5.81 | 1.203832 | 45.6196 | 6.209635 | 24.6033 | 904.981 | 0.0 |
| 7 | 34.0 | 28.5 | 90.0 | 4.5 | 1.0 | 15.2 | 8.1 | 12.3 | 250.0 | 0.0 |
| 8 | 60.0 | 32.1 | 120.0 | 10.0 | 2.5 | 40.0 | 5.5 | 20.0 | 600.0 | 1.0 |
| 9 | 45.0 | 26.7 | 110.0 | 6.5 | 1.6 | 25.3 | 7.0 | 15.0 | 400.0 | 1.0 |
| 10 | 29.0 | 24.3 | 85.0 | 3.0 | 0.7 | 10.5 | 12.0 | 8.0 | 300.0 | 0.0 |
| 11 | 52.0 | 31.2 | 130.0 | 12.0 | 3.0 | 45.0 | 6.0 | 22.0 | 700.0 | 1.0 |
| 12 | 39.0 | 27.8 | 95.0 | 5.5 | 1.2 | 20.0 | 9.0 | 10.0 | 350.0 | 0.0 |
| 13 | 75.0 | 30.48 | 152.0 | 7.01 | 2.628282667 | 50.53 | 10.06 | 11.73 | 99.45 | 1.0 |
| 14 | 60.0 | 26.34929208 | 103.0 | 5.138 | 1.305394533 | 24.2998 | 2.19428 | 20.2535 | 378.996 | 0.0 |
| 15 | 77.0 | 35.58792924 | 76.0 | 3.881 | 0.727558133 | 21.7863 | 8.12555 | 17.2615 | 618.272 | 0.0 |
| 16 | 72.0 | 23.62 | 105.0 | 4.42 | 1.14478 | 21.78 | 17.86 | 4.82 | 195.94 | 1.0 |

## B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 59.0 | 22.83287935 | 98.0 | 6.862 | 1.658774133 | 14.9037 | 4.230105 | 8.2049 | 355.31 | 1.0 |
| 2 | 62.0 | 26.84 | 100.0 | 4.53 | 1.1174 | 12.45 | 21.42 | 7.32 | 330.16 | 1.0 |
| 3 | 57.0 | 34.83814777 | 95.0 | 12.548 | 2.940414667 | 33.1612 | 2.36495 | 9.9542 | 655.834 | 1.0 |
| 4 | 72.0 | 29.13631634 | 83.0 | 10.949 | 2.241625267 | 26.8081 | 2.78491 | 14.76966 | 232.018 | 1.0 |
| 5 | 66.0 | 27.7 | 90.0 | 6.042 | 1.341324 | 24.846 | 7.652055 | 6.7052 | 225.88 | 0.0 |
| 6 | 61.0 | 32.03895937 | 85.0 | 18.077 | 3.790144333 | 30.7729 | 7.780255 | 13.68392 | 444.395 | 0.0 |
| 7 | 34.0 | 27.5 | 85.0 | 5.2 | 1.2 | 15.3 | 8.1 | 10.5 | 300.5 | 0.0 |
| 8 | 45.0 | 30.1 | 92.0 | 14.3 | 3.3 | 25.4 | 7.2 | 12.1 | 450.2 | 1.0 |
| 9 | 50.0 | 28.7 | 100.0 | 8.5 | 1.9 | 18.6 | 9.5 | 11.8 | 375.1 | 0.0 |
| 10 | 39.0 | 32.0 | 95.0 | 6.7 | 1.5 | 20.1 | 6.8 | 9.9 | 410.3 | 1.0 |
| 11 | 54.0 | 29.4 | 110.0 | 12.0 | 2.8 | 22.5 | 5.4 | 13.7 | 520.4 | 1.0 |
| 12 | 67.0 | 31.2 | 88.0 | 9.1 | 2.1 | 19.8 | 8.3 | 10.2 | 390.7 | 0.0 |
| 13 | 77.0 | 35.58792924 | 76.0 | 3.881 | 0.727558133 | 21.7863 | 8.12555 | 17.2615 | 618.272 | 0.0 |
| 14 | 48.0 | 28.125 | 90.0 | 2.54 | 0.56388 | 15.5325 | 10.22231 | 16.11032 | 1698.44 | 1.0 |
| 15 | 58.0 | 29.15451895 | 139.0 | 16.582 | 5.685415067 | 22.8884 | 10.26266 | 13.97399 | 923.886 | 1.0 |
| 16 | 68.0 | 21.36752137 | 77.0 | 3.226 | 0.612724933 | 9.8827 | 7.16956 | 12.766 | 928.22 | 0.0 |

## B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 69.0 | 32.5 | 93.0 | 5.43 | 1.245642 | 15.145 | 11.78796 | 11.78796 | 270.142 | 0.0 |
| 2 | 89.0 | 22.7 | 77.0 | 4.69 | 0.890787333 | 6.964 | 5.589865 | 12.9361 | 1256.083 | 0.0 |
| 3 | 35.0 | 35.2507611 | 90.0 | 6.817 | 1.513374 | 50.6094 | 6.966895 | 22.03703 | 667.928 | 0.0 |
| 4 | 40.0 | 30.83653053 | 128.0 | 41.894 | 13.22733227 | 31.0385 | 6.160995 | 17.55503 | 638.261 | 1.0 |
| 5 | 78.0 | 25.3 | 60.0 | 3.508 | 0.519184 | 6.633 | 10.567295 | 4.6638 | 209.749 | 0.0 |
| 6 | 62.0 | 22.65625 | 92.0 | 3.482 | 0.790181867 | 9.8648 | 11.236235 | 10.69548 | 703.973 | 1.0 |
| 7 | 34.0 | 28.5 | 85.0 | 5.2 | 1.1 | 20.5 | 10.9 | 15.4 | 300.5 | 0.0 |
| 8 | 52.0 | 31.0 | 110.0 | 8.5 | 2.0 | 35.1 | 8.0 | 22.3 | 450.2 | 1.0 |
| 9 | 47.0 | 29.8 | 95.0 | 6.0 | 1.5 | 25.3 | 9.5 | 18.1 | 320.0 | 0.0 |
| 10 | 60.0 | 34.2 | 120.0 | 10.0 | 2.5 | 40.0 | 7.5 | 25.0 | 600.0 | 1.0 |
| 11 | 29.0 | 26.0 | 75.0 | 4.0 | 0.8 | 15.0 | 11.0 | 10.0 | 200.0 | 0.0 |
| 12 | 73.0 | 33.5 | 130.0 | 12.0 | 3.0 | 45.0 | 6.5 | 30.0 | 700.0 | 1.0 |
| 13 | 74.0 | 28.65013774 | 88.0 | 3.012 | 0.6538048 | 31.1233 | 7.65222 | 18.35574 | 572.401 | 1.0 |
| 14 | 65.0 | 32.05 | 97.0 | 5.73 | 1.370998 | 61.48 | 22.54 | 10.33 | 314.05 | 1.0 |
| 15 | 76.0 | 27.2 | 94.0 | 14.07 | 3.262364 | 35.891 | 9.34663 | 8.4156 | 377.227 | 0.0 |
| 16 | 65.0 | 29.666548 | 85.0 | 14.649 | 3.071407 | 26.5166 | 7.28287 | 19.46324 | 1698.44 | 1.0 |

## B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 36.0 | 28.57667585 | 86.0 | 4.345 | 0.921719333 | 15.1248 | 8.6 | 9.1539 | 534.224 | 0.0 |
| 2 | 66.0 | 26.5625 | 89.0 | 6.524 | 1.432235467 | 14.9084 | 8.42996 | 14.91922 | 269.487 | 1.0 |
| 3 | 69.0 | 21.51385851 | 112.0 | 6.683 | 1.846290133 | 32.58 | 4.138025 | 15.69876 | 713.239 | 1.0 |
| 4 | 53.0 | 36.7901662 | 101.0 | 10.175 | 2.534931667 | 27.1841 | 20.03 | 10.26309 | 695.754 | 0.0 |
| 5 | 66.0 | 36.21227888 | 101.0 | 15.533 | 3.869788067 | 74.7069 | 7.53955 | 22.32024 | 864.968 | 0.0 |
| 6 | 75.0 | 30.48 | 152.0 | 7.01 | 2.628282667 | 50.53 | 10.06 | 11.73 | 99.45 | 1.0 |
| 7 | 34.0 | 27.5 | 85.0 | 5.2 | 1.1 | 20.5 | 10.2 | 12.3 | 400.5 | 0.0 |
| 8 | 58.0 | 31.2 | 110.0 | 8.5 | 2.0 | 45.2 | 6.8 | 14.5 | 600.3 | 1.0 |
| 9 | 42.0 | 29.8 | 95.0 | 7.0 | 1.5 | 30.1 | 9.5 | 11.0 | 500.8 | 0.0 |
| 10 | 65.0 | 34.1 | 120.0 | 12.0 | 3.5 | 60.0 | 5.0 | 18.0 | 750.2 | 1.0 |
| 11 | 50.0 | 25.0 | 90.0 | 4.5 | 0.9 | 15.0 | 12.0 | 10.0 | 300.0 | 0.0 |
| 12 | 39.0 | 28.0 | 88.0 | 6.0 | 1.2 | 22.0 | 11.0 | 13.0 | 450.1 | 0.0 |
| 13 | 35.0 | 30.27681661 | 84.0 | 4.376 | 0.9067072 | 39.2134 | 9.048185 | 16.43706 | 733.797 | 0.0 |
| 14 | 69.0 | 32.5 | 93.0 | 5.43 | 1.245642 | 15.145 | 11.78796 | 11.78796 | 270.142 | 0.0 |
| 15 | 58.0 | 29.15451895 | 139.0 | 16.582 | 5.685415067 | 22.8884 | 10.26266 | 13.97399 | 923.886 | 1.0 |
| 16 | 49.0 | 29.77777778 | 70.0 | 8.396 | 1.449709333 | 51.3387 | 10.73174 | 20.76801 | 602.486 | 1.0 |

## B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 44.0 | 19.56 | 114.0 | 15.89 | 4.468268 | 13.08 | 20.37 | 4.62 | 220.66 | 1.0 |
| 2 | 51.0 | 18.37 | 105.0 | 6.03 | 1.56177 | 9.62 | 12.76 | 3.21 | 513.66 | 1.0 |
| 3 | 60.0 | 31.23140988 | 131.0 | 30.13 | 9.736007333 | 37.843 | 8.40443 | 11.50005 | 396.021 | 1.0 |
| 4 | 89.0 | 22.7 | 77.0 | 4.69 | 0.890787333 | 6.964 | 5.589865 | 12.9361 | 1256.083 | 0.0 |
| 5 | 42.0 | 21.35991456 | 93.0 | 2.999 | 0.6879706 | 19.0826 | 8.462915 | 17.37615 | 321.919 | 1.0 |
| 6 | 38.0 | 22.4996371 | 95.0 | 5.261 | 1.232827667 | 8.438 | 4.77192 | 15.73606 | 199.055 | 1.0 |
| 7 | 34.0 | 25.4 | 90.0 | 8.5 | 2.0 | 15.2 | 10.5 | 12.3 | 300.5 | 0.0 |
| 8 | 45.0 | 28.7 | 110.0 | 12.0 | 3.5 | 20.1 | 9.8 | 14.0 | 450.0 | 1.0 |
| 9 | 52.0 | 31.2 | 120.0 | 15.0 | 4.0 | 25.0 | 8.5 | 16.0 | 600.0 | 1.0 |
| 10 | 39.0 | 22.5 | 95.0 | 6.5 | 1.5 | 12.0 | 11.0 | 10.5 | 250.0 | 0.0 |
| 11 | 60.0 | 29.9 | 130.0 | 20.0 | 5.0 | 30.0 | 7.0 | 18.0 | 700.0 | 1.0 |
| 12 | 48.0 | 27.3 | 100.0 | 9.0 | 2.5 | 22.5 | 9.5 | 13.0 | 400.0 | 1.0 |
| 13 | 45.0 | 37.03560819 | 83.0 | 6.76 | 1.383997333 | 39.9802 | 4.617125 | 8.70448 | 586.173 | 0.0 |
| 14 | 61.0 | 32.03895937 | 85.0 | 18.077 | 3.790144333 | 30.7729 | 7.780255 | 13.68392 | 444.395 | 0.0 |
| 15 | 46.0 | 22.21 | 86.0 | 36.94 | 7.836205333 | 10.16 | 9.76 | 5.68 | 312.0 | 1.0 |
| 16 | 85.0 | 27.68877813 | 196.0 | 51.814 | 25.05034187 | 70.8824 | 7.901685 | 55.2153 | 1078.359 | 1.0 |

## B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 86.0 | 26.66666667 | 201.0 | 41.611 | 20.6307338 | 47.647 | 5.357135 | 24.3701 | 1698.44 | 1.0 |
| 2 | 43.0 | 34.42217362 | 89.0 | 23.194 | 5.091856133 | 31.2128 | 8.300955 | 6.71026 | 960.246 | 0.0 |
| 3 | 59.0 | 22.83287935 | 98.0 | 6.862 | 1.658774133 | 14.9037 | 4.230105 | 8.2049 | 355.31 | 1.0 |
| 4 | 68.0 | 35.56 | 131.0 | 8.15 | 2.633536667 | 17.87 | 11.9 | 4.19 | 198.4 | 1.0 |
| 5 | 62.0 | 26.84 | 100.0 | 4.53 | 1.1174 | 12.45 | 21.42 | 7.32 | 330.16 | 1.0 |
| 6 | 83.0 | 20.69049454 | 92.0 | 3.115 | 0.706897333 | 8.8438 | 5.429285 | 4.06405 | 468.786 | 0.0 |
| 7 | 34.0 | 28.5 | 110.0 | 8.5 | 1.9 | 25.3 | 10.5 | 15.2 | 400.1 | 0.0 |
| 8 | 57.0 | 31.2 | 150.0 | 22.0 | 5.0 | 30.1 | 9.8 | 12.3 | 600.5 | 1.0 |
| 9 | 45.0 | 24.0 | 95.0 | 6.0 | 1.4 | 20.0 | 15.0 | 8.5 | 300.2 | 0.0 |
| 10 | 62.0 | 29.8 | 130.0 | 15.0 | 3.5 | 28.0 | 12.0 | 10.0 | 500.0 | 1.0 |
| 11 | 39.0 | 26.5 | 105.0 | 7.5 | 1.8 | 22.5 | 11.0 | 14.5 | 350.3 | 0.0 |
| 12 | 50.0 | 33.0 | 140.0 | 18.0 | 4.0 | 35.0 | 8.0 | 16.0 | 450.7 | 1.0 |
| 13 | 78.0 | 25.3 | 60.0 | 3.508 | 0.519184 | 6.633 | 10.567295 | 4.6638 | 209.749 | 0.0 |
| 14 | 52.0 | 30.8012487 | 87.0 | 30.212 | 6.4834952 | 29.2739 | 6.26854 | 24.24591 | 764.667 | 1.0 |
| 15 | 45.0 | 20.26 | 92.0 | 3.44 | 0.780650667 | 7.65 | 16.67 | 7.84 | 193.87 | 1.0 |
| 16 | 59.0 | 28.67262608 | 77.0 | 3.188 | 0.605507467 | 17.022 | 16.44048 | 31.6904 | 910.489 | 1.0 |

## B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 51.0 | 22.89281998 | 103.0 | 2.74 | 0.696142667 | 8.0163 | 9.349775 | 11.55492 | 359.232 | 1.0 |
| 2 | 48.0 | 31.25 | 199.0 | 12.162 | 5.9699204 | 18.1314 | 4.104105 | 53.6308 | 1698.44 | 1.0 |
| 3 | 36.0 | 28.57667585 | 86.0 | 4.345 | 0.921719333 | 15.1248 | 8.6 | 9.1539 | 534.224 | 0.0 |
| 4 | 34.0 | 21.47 | 78.0 | 3.469 | 0.6674356 | 14.57 | 13.11 | 6.92 | 354.6 | 0.0 |
| 5 | 66.0 | 36.21227888 | 101.0 | 15.533 | 3.869788067 | 74.7069 | 7.53955 | 22.32024 | 864.968 | 0.0 |
| 6 | 24.0 | 18.67 | 88.0 | 6.107 | 1.33 | 8.88 | 36.06 | 6.85 | 632.22 | 0.0 |
| 7 | 29.0 | 24.5 | 90.0 | 5.1 | 1.1 | 12.5 | 8.0 | 10.0 | 400.0 | 0.0 |
| 8 | 54.0 | 32.0 | 150.0 | 8.0 | 2.0 | 30.0 | 6.5 | 15.0 | 600.0 | 1.0 |
| 9 | 38.0 | 27.3 | 110.0 | 7.5 | 1.5 | 15.0 | 9.0 | 12.0 | 500.0 | 0.0 |
| 10 | 45.0 | 29.8 | 120.0 | 9.0 | 2.3 | 20.0 | 7.5 | 14.0 | 550.0 | 1.0 |
| 11 | 60.0 | 35.0 | 140.0 | 10.5 | 3.0 | 25.0 | 5.0 | 18.0 | 700.0 | 1.0 |
| 12 | 33.0 | 22.0 | 85.0 | 4.0 | 0.8 | 10.0 | 11.0 | 8.0 | 300.0 | 0.0 |
| 13 | 64.0 | 34.5297228 | 95.0 | 4.427 | 1.037393667 | 21.2117 | 5.46262 | 6.70188 | 252.449 | 0.0 |
| 14 | 32.0 | 36.51263743 | 87.0 | 14.026 | 3.0099796 | 49.3727 | 5.1 | 17.10223 | 588.46 | 0.0 |
| 15 | 43.0 | 31.25 | 103.0 | 4.328 | 1.099600533 | 25.7816 | 12.71896 | 38.6531 | 775.322 | 1.0 |
| 16 | 57.0 | 34.83814777 | 95.0 | 12.548 | 2.940414667 | 33.1612 | 2.36495 | 9.9542 | 655.834 | 1.0 |